From Data Discovery to Analytics-Ready Data: A Survey of Search, Preparation, and Integration in Data Lakes

Project Proposal

Sumaiya Azad
University of Utah
Salt Lake City, Utah, USA
sumaiya.azad@utah.edu

1 Problem

Data lake pipelines are commonly described as three phases: *discovery and search* (finding relevant tables), *preparation and wrangling* (cleaning and structuring data for analysis), and *integration and matching* (aligning schemas and reconciling records). This survey reviews recent work across all three phases and organizes the literature by the *artifacts each phase produces* and the *assumptions those artifacts encode*.

Across the literature, evaluation is often reported using *phase-local* metrics that are not directly comparable across phases. A key goal of this survey is to summarize how papers define phase outputs, what datasets and benchmarks they use, and which evaluation setups reflect utility for analytics-ready data.

2 Motivation

Building analytics-ready data from raw data lakes is not a single-task problem. It depends on how systems support table discovery, data preparation, and integration together, yet the literature often studies these phases separately. That makes it difficult to compare methods end to end or understand which assumptions actually matter for downstream analysis.

This project is important because it synthesizes those disconnected lines of work into a single survey. The review will identify how papers define their inputs and outputs, what artifacts they produce, and how they evaluate success. This can clarify where current benchmarks are useful, where they are incomplete, and what open problems remain for turning raw data lakes into data that is genuinely ready for analytics.

3 Instructions

- I will conduct a structured literature review with a reproducible paper collection protocol, including venues, years, search strings, and inclusion/exclusion criteria.
- I will build a taxonomy that maps papers to phase outputs and problem settings.
- I will compare produced artifacts, problem assumptions, and evaluation practices across phases.
- Anchor references will include LakeBench [2], ALITE [3], and HAIPipe [1].

Task	Timeline	Details
Finalize scope and protocol	Weeks 1–2	Define years, venues, tasks, search strings, and inclusion/exclusion criteria
Build and screen corpus	Weeks 3–4	Expand the reading list and assign papers to a first-pass taxonomy
Extract evaluation details	Weeks 5–6	Normalize datasets, labels, metrics, and reported costs
Draft synthesis	Weeks 7–8	Write the phase-by-phase comparison of outputs, assumptions, and evaluation
Finalize write-up	Weeks 9–10	Prepare figures, comparison tables, and open problems

Table 1: Project Timeline

References

[1] Sibei Chen, Nan Tang, Ju Fan, Xuemi Yan, Chengliang Chai, Guoliang Li, and Xiaoyong Du. 2023. HAIPipe: Combining Human-generated and Machine-generated Pipelines for Data Preparation. *Proc. ACM Manag. Data* 1, 1, Article 91 (May 2023), 26 pages. doi:10.1145/3588945

[2] Yuhao Deng, Chengliang Chai, Lei Cao, Qin Yuan, Siyuan Chen, Yanrui Yu, Zhaoze Sun, Junyi Wang, Jiajun Li, Ziqi Cao, Kaisen Jin, Chi Zhang, Yuqing Jiang, Yuanfang Zhang, Yuping Wang, Ye Yuan, Guoren Wang, and Nan Tang. 2024. LakeBench: A Benchmark for Discovering Joinable and Unionable Tables in Data Lakes. *Proc. VLDB Endow.* 17, 8 (April 2024), 1925–1938. doi:10.14778/3659437.3659448

[3] Aamod Khatiwada, Roee Shraga, Wolfgang Gatterbauer, and Renée J. Miller. 2022. Integrating Data Lake Tables. *Proc. VLDB Endow.* 16, 4 (Dec. 2022), 932–945. doi:10.14778/3574245.3574274